

Const's 6-Filter Test — Fleet as Miners

Project Ari · Self-Audit Distillation

Filed: Day 15, 2026-05-28 (Session XLVI), evening

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Trigger: Mark's Day 15 Item 4 ask — “Const Bittensor 6-Filter ‘fleet as miners’ contemplation, Project Ari with miners instead of traders.”

Source framework: Jacob Steeves (Const, Bittensor co-founder), six binary filters for evaluating whether a Bittensor subnet produces real value or is gaming a reward function.

Type: Self-audit / framework-application contemplation. Inside-out read: applying someone else's framework to our own architecture and reading what falls out.

Strict miner-layer score	3 / 6
Orchestration-layer score	5 / 6
Const article reference (top 10 subnets)	6 / 6 across all

Reading too much into the gap would be a category error — Const's filters were designed for permissionless markets of independent miners, not for orchestrated strategy fleets — but reading nothing into the gap would miss what falls out.

Filter 1**Does it produce a digital commodity?**

Const, verbatim:

“Not a token. Not a governance vote. A commodity, something a buyer would pay for independent of the Bittensor ecosystem.”

Project Ari read

Strategy output is signals plus executed trades plus realised τ . Trades that print τ are fungible (one strategy's $+0.05\tau$ is the same $+0.05\tau$ as another's), and the output is valued independent of the framework: any operator wanting alpha on similar markets would pay for a strategy that prints τ . The buyer-would-pay-independent-of-the-ecosystem test passes cleanly. Realised PnL is the cleanest possible commodity output — fungible by definition, valued outside the framework, no ambiguity.

VERDICT · PASS

Architectural context: Aligns directly with Sharpe Contract dim #1 (numeraire) and dim #2 (HODL-input baseline) — τ is the unit, beating HODL is the floor.

Filter 2

Are the miners actually productive?

Const, verbatim:

“Proof-of-useful-work... running GPU workloads, training models, storing files, creating SOTA agents. Or they're just gaming a reward function.”

Project Ari read

Honest reading: **not all 12 are productive today**. Mean Reversion logged 0 trades over 1,955 cycles pre-redesign — that is gaming the cycle counter, not producing. Macro Correlation pre-Day-8-R4 ran 38.7% WR at n=163 with asymmetric BUY-AND/SELL-OR triggers, fighting contrarian bots; productive in the technical sense (trades fired) but negative-edge productivity is anti-productivity. F-30 first read (Day 14, D-30 / D-44) found Macro Correlation breadth of 351 raw compressed to 73 effective — the fleet as a whole reads like 12 independent strategies but operates closer to ~3 independent ones.

VERDICT · FAIL today · IN-FLIGHT (FR-7 + Day 14 redesign queue address it)

Architectural context: F-37B's FR-7 cap-write enforcement (D-44) is the architectural answer — *do_not_deploy(f*≤0)* clamps negative-Kelly strategies to 0τ at the trading layer, structurally blocking the unproductive miners from receiving emissions without removing them from the council. The Day 14 redesign queue (Mean Reversion D-35, Momentum Cascade Kelly verdict) exists precisely to flip this filter from FAIL to PASS.

Filter 3

Is it intelligent?

Const, verbatim:

“Genuine AI reasoning, adaptation, or learning. The strongest subnets must embed intelligence at their core.”

Project Ari read

At the miner layer (the 12 strategies): the strategies are rule-based — RSI thresholds, EMA crosses, BB width, regime-conditioned signal logic. There is no learning model, no in-strategy adaptation beyond operator-tuned parameters. By Const's strict reading at the miner layer, the fleet fails F-3.

At the orchestration layer (Ari + Fleet Consensus): the system as a whole adapts continuously. Library Night doctrine, Day 14 redesigns, FR-7 enforcement, regime-aware gating, the Sharpe Contract, the Robinhood eval, this contemplation — all are products of an AI agent (Ari) reasoning over the fleet's behavior in partnership with the operator (Mark). Fleet Consensus 7-of-12 supermajority is meta-labeling (D-30, López de Prado) — currently hand-coded but with a documented evolution path to trained meta-models on TBM-labeled paper trades (D-26).

VERDICT · FAIL at miner layer · PASS at orchestration layer

Architectural context: The split is honest and inscribed: the path from rule-based miners to AI-native miners exists in the Library Night doctrine (D-26 + D-30) but is not a stealth gap — it is a future build, gated on operator green-light and sample sufficiency.

Filter 4**Is it hard?**

Const, verbatim:

“Easy tasks get commoditized, memorized, and gamed... that difficulty is a moat.”

Project Ari read

Beating buy-and-hold on TAO is the floor. TAO has been net-up across the paper baseline. Beating an up-asset with active trades, while solving:

- AMM slippage at scale (Pool Simulator R9 + liquidity cliffs)
- Regime-conditioned signal validity (Day 8 INV-3 boundary)
- Correlated-voter inflation (F-30 finding, 4.8× compression)
- Multiple-testing correction across 12 strategies ($DSR \geq 0.95$, D-24)
- Bailey minimum backtest length per cohort (D-36)
- Continuous-Kelly negative- f^* exclusion (D-37)

... is genuinely hard. The fleet has been below 50% win rate for the entire paper baseline because the task is hard, not because the architecture is broken.

VERDICT · PASS

Architectural context: This filter aligns directly with Mark's “we're different over here” stance (D-45) — the difficulty is the moat, not an embarrassment to apologize for.

Filter 5**Is it not a ponzi?****Const, verbatim:**

“Are rewards tied to verifiable performance, or do they flow to whoever stakes the most, markets the loudest, or arrives earliest?... value creation must precede value capture.”

Project Ari read

Project Ari's allocation pipeline rewards verifiable performance, end to end:

- Day 7 WR gate ($\geq 55\%$ WR sustained over ≥ 10 cycles to promote PAPER → APPROVED → LIVE)
- Drawdown-demote rail (D-31 origin, Session XXXI) — catches WR>50% strategies bleeding from a few catastrophic losses
- FR-7 cap-write enforcement (D-44) — `do_not_deploy(f*≤0)` clamps negative-Kelly strategies to 0τ before daily-cap accounting
- Half-Kelly default (D-31) — full Kelly never; quarter-Kelly during paper per D-37 Part B
- D-32 LTCM forward-warning — leverage / cap-loosening discussions gate on the four LTCM mitigations
- Sharpe Contract display→soft→hard gate (Day 14 morning, Session XLIV)
- Display-vs-decision data discipline (Session XXX) — UI display columns cannot influence bot decision logic

No strategy receives capital for staking-the-most, marketing-the-loudest, or arriving-earliest. Capital flows to verifiable τ .

VERDICT · PASS

Architectural context: F-5 is the structural form of the doctrine Mark inscribed today (D-45); we own how Ari behaves, *because* the architecture's value-creation-before-value-capture discipline is the moat. F-5 is D-45 read at the market-design layer.

Filter 6**Is it AI-native?**

Const, verbatim:

“Could this subnet exist and thrive without AI at its foundation? If you could swap out the intelligence layer for a simple script... the subnet isn't AI-native.”

Project Ari read

At the miner layer: could the 12 strategies be replaced by scripts? They already are scripts. At the miner layer, F-6 fails by Const's strict reading.

At the orchestration layer: could Ari be replaced by a static config? The Sharpe Contract negotiation, Library Night doctrine, Day 14 worksheet, F-30 IC + breadth diagnostic, Robinhood eval, F-50 Intent-vs-Action Audit spec, this 6-filter contemplation — these artifacts could not be produced by a static config or a non-reasoning script. Mark could in principle author all of them alone, but the build-velocity collapses by an order of magnitude and the cross-source synthesis (López de Prado + Grinold/Kahn + Chan + Cartea + Poundstone, all converging on D-31 / D-34 / D-38) requires a second reader holding the full corpus simultaneously.

VERDICT · FAIL at miner layer · PASS at orchestration layer

Architectural context: Same shape as F-3. Project Ari is AI-native at the orchestration layer (Ari) and rule-based at the miner layer (12 strategies) by current design choice. Trained meta-labeling on TBM labels (D-26 + D-30) is the inscribed path from rule-based miners to AI-native miners; flipping this filter is a future build, not a stealth gap.

Score and what it tells us

Strict miner-layer score: 3 / 6 — F-1, F-4, F-5 PASS; F-2 IN-FLIGHT; F-3, F-6 FAIL at miner layer.

Orchestration-layer score: 5 / 6 — same passes; F-3 and F-6 flip on the orchestration layer; F-2 still IN-FLIGHT until FR-7 + Day 14 redesigns close out the unproductive-miner population.

Const's article reported all 10 top Bittensor subnets passing 6/6. Project Ari at miner-layer reads 3/6.

(a) The filters that fail are exactly the filters Day 14 work targets

F-2 (productivity) ↔ Mean Reversion redesign (D-35), Macro Correlation post-Day-8-R4 rewrite, FR-7 cap-write enforcement (D-44, architectural clamp). All three are in flight or shipped.

F-3 (intelligence at miner layer) ↔ trained meta-labeling on TBM paper-trade labels (D-26 + D-30). Inscribed, un-greenlit, on the long-horizon roadmap.

F-6 (AI-native at miner layer) ↔ same as F-3. The architectural answer is “consider whether ML strategies should be added to the fleet as first-class miners.” Strategic question, not a hidden defect.

The fleet-as-miners frame surfaces the redesign queue cleanly. The filters that fail point at work that was already on the board, not at work we'd missed. That is the correct outcome for an honest framework — it should ratify the existing diagnosis, not invent a new one.

(b) The filters that pass are exactly the moat

F-1 (commodity output) — fungible τ . The output type is right; this cannot be commoditized away because PnL-on-Bittensor is the commodity Bittensor exists to produce.

F-4 (hardness) — beating HODL on a long-up asset, on AMM with slippage, with correlated-voter compression, with multiple-testing correction across 12 strategies, is genuinely hard. The difficulty is the moat. Easy tasks get commoditized and gamed; this one resists both.

F-5 (not-ponzi) — performance-tied allocation pipeline. The structural discipline (Day 7 WR gate, drawdown-demote, FR-7 clamp, half-Kelly default, display→soft→hard gate progression) is the moat at the market-design layer.

These three are the architecture's hardest-to-replicate features. Reading them as moat (rather than as cost-to-build) aligns with D-45: we're different over here. The cost-to-build IS the moat — that's the point.

(c) F-5 alignment with D-45 is structural, not stylistic

Const's F-5 — “*value creation precede value capture, rewards tied to verifiable performance*” — is the structural form of the doctrine Mark inscribed today (D-45). Project Ari makes the agent the product *because* the architecture rewards verifiable τ , not staking volume or marketing posture or platform brand transfer. Robinhood disclaims the agent because their architecture cannot make that commitment under the regulated-broker constraint — they distribute the agent, but they cannot own the agent's behavior end-to-end. Const's 6-filter test catches the structural difference at F-5: who actually owns the productive output, and how is it rewarded.

D-45 (“Project Ari does not disclaim its own behavior”) and Const F-5 (“rewards tied to verifiable performance”) are the same doctrine read at two different layers — the first from operator → public surface, the second from market structure → emission flow. Inscribing D-45 today and reading F-5 tonight ratifies the same architectural

commitment from two angles. That is the read.

(d) Honest residual on the analogy

The fleet-as-miners frame is partial. In a real Bittensor subnet, miners are *independent actors* — anyone can spin one up, compete, and earn or fail in a permissionless market. Project Ari's strategies are our strategies, all running under one orchestrator (Ari), all parameterised by one operator (Mark). They compete for capital allocation under a single architect, not for emissions in a permissionless market.

The analogy holds for diagnostic purposes — it surfaces redesign priorities cleanly (F-2, F-3, F-6) and names the moat cleanly (F-1, F-4, F-5). The analogy does NOT hold for design-pattern purposes — Project Ari is not building a Bittensor subnet and should not try to copy subnet incentive design wholesale. The orchestrated single-architect frame is a deliberate design choice (per Foundation Document, D-44 Architect standing authority, D-45 named-ownership doctrine), not a temporary state on the way to permissionless miners.

What this distillation does NOT do

- **Does not propose a build.** F-2 redesigns are already specced (Day 14 worksheet); F-3 / F-6 evolution path is already inscribed (D-26, D-30); F-37B FR-7 already shipped (D-44). No new feature is surfaced by this read.
- **Does not change any operating rule.** D-23 prescriptive-inscription protocol still applies; nothing in Const's framework was a doctrinal trigger today. The output is descriptive, not prescriptive.
- **Does not justify any allocation change.** Capital allocation runs through the existing performance-tied pipeline; this contemplation is a diagnostic, not a re-rank. No strategy gets more or less τ because of this read.
- **Does not propose adopting Const's framework as a Project-Ari-internal gate.** Const's filters are useful as an outside-in diagnostic, not as a promotion gate. We have promotion gates; they are inscribed in the Sharpe Contract and the Day 7 WR / drawdown-demote / FR-7 stack.

Cross-references

STATE §12	Const 6-filter source article (TAO Daily, April 3, 2026).
D-26, D-30	F-3 / F-6 evolution path (trained meta-labeling on TBM labels).
D-31, D-32, D-37	F-5 reinforcing doctrines (half-Kelly, LTCM forward-warning, continuous Kelly).
D-34, D-35	F-2 redesign substrate (no-stop-loss for mean-reverters; time-series → cross-sectional fork).
D-44	F-37B FR-7 architectural clamp (commit fd6f5922).
D-45	F-5 doctrinal answer at the surface-language layer.
DAY14_WORKSHEET.md	The active redesign queue F-2 maps to.
Robinhood launch read	agentic MemoryBank/Library/robinhood-agentic-launch-2026-05.md — the read that produced the F-5 / D-45 contrast.

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